AICE2009 Lab 3

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1 Linear Time-Invariant Systems

A Linear Time-Invariant state-space system is defined by linearity (it obeys the principles of superposition) and time-invariance (its parameters do not change over time). Because of these properties, an LTI system can be modeled using constant matrices (e.g., A and B). When a system is nonlinear, its states or inputs interact in ways that violate superposition, meaning it cannot be exactly represented by constant matrices and must be linearized.

Derivation of the state-space equations: From Newton's second law, the force balance is $m\ddot{x}(t) = u(t) - b\dot{x}(t)$. We define the state vector using position and velocity: $x_1 = x$ and $x_2 = v = \dot{x}$. Rearranging the equation yields the following first-order derivatives:

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= \frac{1}{m}u(t) - \frac{b}{m}x_2\end{aligned}$$

Writing this in matrix form $\dot{x} = Ax + Bu$:

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ 0 & -b/m\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ 1/m\end{bmatrix}u(t) \tag{1}$$

Assuming both position and velocity are measured, the output equation $y = Cx$ is:

$$y = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} \tag{2}$$

2 Kalman Filter

The matrix P_k represents the state estimation error covariance. The diagonal elements of this matrix are the variances of the individual state errors. Therefore, minimizing the trace of P_k physically means minimizing the total sum of the mean squared estimation errors for all states in the system.

The term $E[e_{k-1}v_k^\top]$ represents the expected value of the correlation between the previous step's estimation error (e_{k-1}) and the current measurement noise (v_k). It equals zero because the measurement noise at the current timestep is completely independent of the system's past state errors. For this to hold, we assume that the measurement noise v_k is zero-mean, and completely independent of the process noise and initial state.

The measurement update reduces estimation covariance because adding a new piece of information to a mathematical prediction mathematically reduces overall uncertainty. However, the covariance does not decrease monotonically to zero because process noise (Q) is added during every prediction step to account for physical disturbances, or mathematical approximations.

3 Extended Kalman Filter (EKF)

Using the a posteriori state update, the a posteriori error is $e_k^+ = (I - K_k C_k) e_k^- - K_k v_k$. Taking the expected value $E[e_k^+ (e_k^+)^\top]$ yields the Joseph form:

$$P_k^+ = (I - K_k C_k) P_k^- (I - K_k C_k)^\top + K_k R_k K_k^\top \tag{3}$$

By substituting the optimal Kalman gain $K_k = P_k^- C_k^\top (C_k P_k^- C_k^\top + R_k)^{-1}$ and factoring, this simplifies to the standard update $P_k^+ = (I - K_k C_k) P_k^-$. The Joseph form is highly preferred in numerical implementations because it mathematically guarantees that the covariance matrix P_k^+ remains symmetric and positive semi-definite.

For the continuous-time dynamics $x = [\theta, \omega]^\top$:

- $f(x, u) = \begin{bmatrix} \omega \\ -\frac{g}{\ell} \sin\theta \end{bmatrix}$
- $c(x) = \sin(\theta)$
- $F_k = \frac{\partial f}{\partial x} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{\ell} \cos\theta & 0 \end{bmatrix}$
- $C_k = \frac{\partial c}{\partial x} = \begin{bmatrix} \cos\theta & 0 \end{bmatrix}$

The EKF uses Taylor series expansion to linearize a nonlinear system at the current operating point. If the system is strictly linear to begin with, the partial derivative evaluate exactly to the constant matrices A and C. Under these conditions, the EKF equations reduce to the standard linear Kalman filter equations.

To study the effect of process noise (Q) and measurement noise (R), the EKF was simulated using a deliberately poor initial state guess ($\hat{\theta}_0 = 0.3$ rad vs. true $\theta_0 = 2.8$ rad) under two configurations:

By setting P_0 and Q to extremely small values (e.g., 10^{-10}) and R to a very high value (e.g., 10.0), the filter is forced to trust its flawed initial state and physical model while completely ignoring the sensor data. Consequently, the estimate diverges and completely fails to track the true downward trajectory.

By setting a high initial uncertainty ($P_0 = 1.0$), a reasonable process noise ($Q = 10^{-3}$), and using the true measurement noise variance ($R = 0.05^2$), the filter acknowledges its initial error and heavily weights the incoming sensor measurements. As a result, the filter's estimate quickly converges to the true trajectory and tracks it afterward.

In essence, the Q and R matrices control how much the filter relies on the system model versus the sensor measurements. If Q is much smaller than R, the filter becomes overly confident in its predictions and fails to adapt to new data. A properly tuned EKF balances these two matrices.

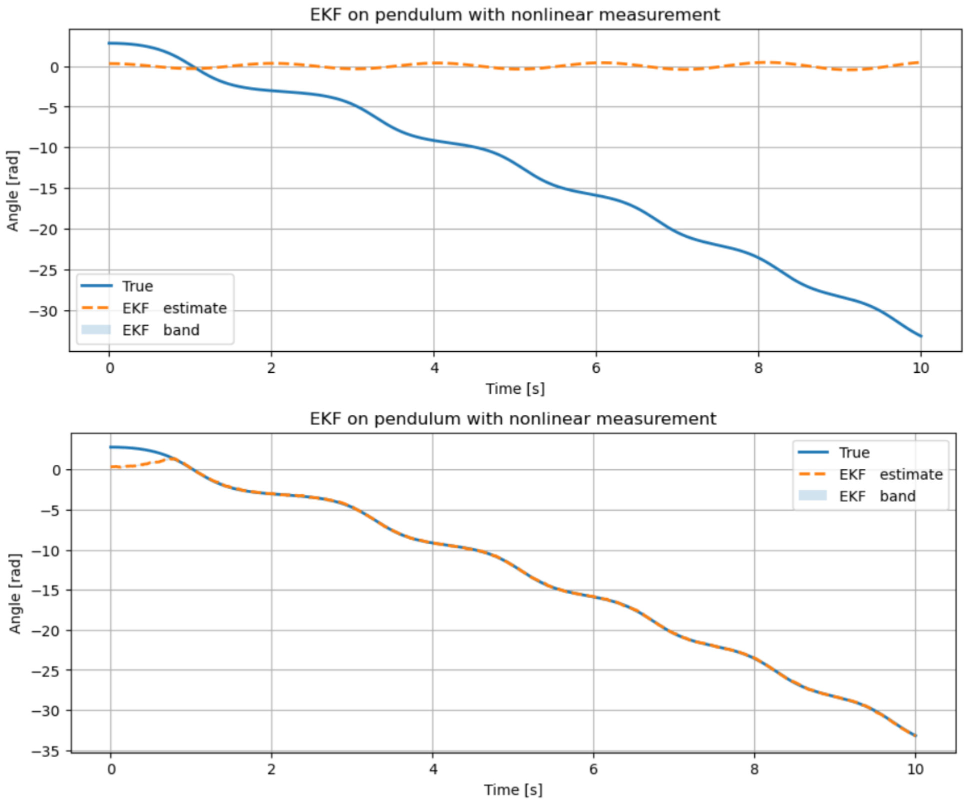

Figure 1: EKF state estimation: Overconfident filter (top) ignoring measurements vs. Well-tuned filter (bottom) rapidly converging to the true state.